

Topic 4: Encapsulation and Access Specifiers

4.1 The Concept of Encapsulation

- Encapsulation means bundling data and functions that operate on that data inside one class, and hiding internal details from the outside
- Protects data from unintended modification -- prevents invalid states
- Users of the class only need to know what it does, not how it does it (black-box principle)
- Achieved in C++ using private data members and public getter/setter functions

4.2 Access Specifiers

- public: accessible from anywhere -- inside the class, derived classes, and outside code
- private: accessible only within the class itself -- not even derived classes (default for class)
- protected: accessible within the class and its derived classes -- covered in inheritance
- A class can have multiple public and private sections in any order

```
class BankAccount {
private:
    double balance;
    string owner;

public:
    BankAccount(string o, double b) : owner(o), balance(b) {}

    void deposit(double amount) {
        if (amount > 0) balance += amount;
    }
    bool withdraw(double amount) {
        if (amount > balance) return false;
        balance -= amount; return true;
    }
    double getBalance() const { return balance; }
};
```

4.3 Getters and Setters

- Getter (accessor): returns the value of a private member -- conventionally named getXxx()
- Setter (mutator): sets the value after validation -- conventionally named setXxx()
- Allows you to add validation logic and change internal representation without breaking external code
- Mark getters as const to signal they do not modify the object

```
class Temperature {
    double celsius;
public:
    void setCelsius(double c) {
        if (c < -273.15) return; // validation
        celsius = c;
    }
    double getCelsius() const { return celsius; }
    double getFahrenheit() const { return celsius * 9/5 + 32; }
};
```

4.4 const Member Functions

- Declaring a member function const (void show() const) promises it will not modify any data member
 - Required when calling functions on const objects
 - Helps catch accidental mutations at compile time -- mark every getter as const
-